R1: Download + Intro to R & Rstudio

Meike Niederhausen and Nicky Wakim

2025-10-06

Introduction to R

Artwork by @allison_horst

What is R?

- A programming language
- Focus on statistical modeling and data analysis
- Useful for epidemiology, biostatistics, and data science
- Great visualizations

What is RStudio?

R: Engine	RStudio: Dashboard
	

Modern Dive

- R is a programming language
- RStudio is an integrated development environment (IDE)
  - An interface to use R (with perks!)

We open RStudio on our computer (not R!)

1.1.2 Using R via RStudio

Recall our car analogy from earlier. Much as we don't drive a car by interacting directly with the engine but rather by interacting with elements on the car's dashboard, we won't be using R directly but rather we will use RStudio's interface. After you install R and RStudio on your computer, you'll have two new *programs* (also called *applications*) you can open. We'll always work in RStudio and not in the R application. Figure 1.2 shows what icon you should be clicking on your computer.

R: Do not open this	RStudio: Open this

FIGURE 1.2: Icons of R versus RStudio on your computer.

Modern Dive

If you have issues downloading on your laptop for in-class use

- There is a really cool online interface that will be sufficient for our in-class use!
- https://posit.cloud/
  - If you Google "Posit cloud" it should come up easily
- I don't recommend this for homework use (once we get to it)

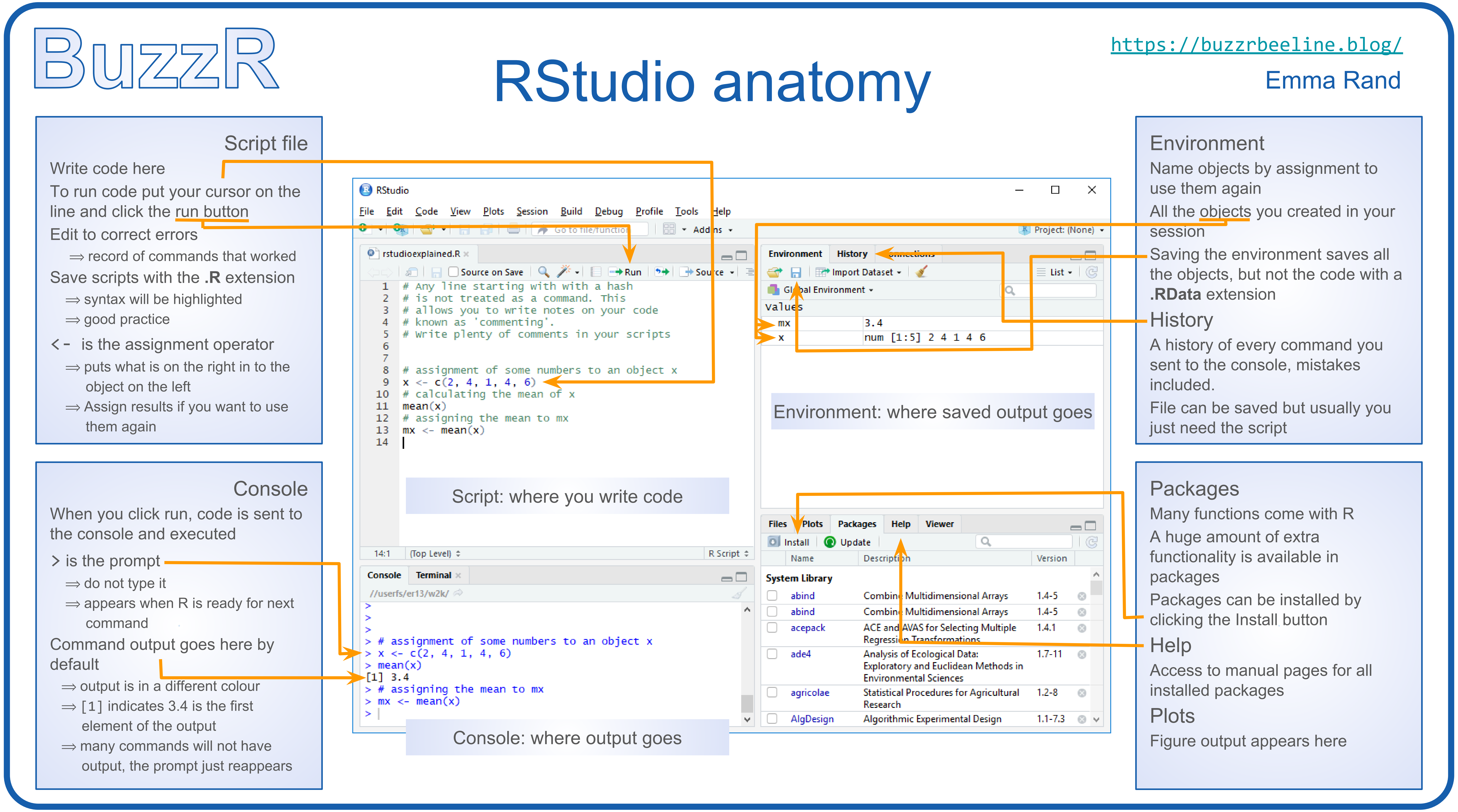
So let's take some time to download each!

- Use this link to start: https://posit.co/download/rstudio-desktop/
- I will click on this link!

- You must install R first
  - Even if you have R installed already, I highly recommend installing the latest version
  - In the future, you will periodically want to update this
- Install RStudio Desktop Open Source License (second)

- If you get them both installed, then you can open up RStudio and start checking it out

RStudio anatomy



Emma Rand